CSCI3230 / ESTR3108 2026–27 First Term Assignment 2

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University Guideline on Academic Honesty:
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`http://www.erg.cuhk.edu.hk/erg-intra/upload/documents/ENGG_Discipline.pdf`

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Question 1: SVM

(a) Dual problem, support vectors, hyperplane

Dual problem:

$$\max_{\boldsymbol{\alpha}} \sum_{i=1}^{8} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{8} \alpha_i \alpha_j y_i y_j \, \mathbf{x}_i^\top \mathbf{x}_j \qquad \text{s.t.} \quad \sum_{i=1}^{8} \alpha_i y_i = 0, \ \alpha_i \geq 0.$$

Since $\alpha_1 = \alpha_8 = 0.8 > 0$ (others 0), by KKT the support vectors are **point 1** $(3.5, -2, +1)$ **and point 8** $(4, -3.5, -1)$.

$$\mathbf{w}^* = \sum_i \alpha_i y_i \mathbf{x}_i = 0.8 \begin{pmatrix} 3.5 \\ -2 \end{pmatrix} - 0.8 \begin{pmatrix} 4 \\ -3.5 \end{pmatrix} = \begin{pmatrix} -0.4 \\ 1.2 \end{pmatrix},$$

$$b^* = y_1 - \mathbf{w}^{*\top} \mathbf{x}_1 = 1 - \big[(-0.4)(3.5) + (1.2)(-2)\big] = 4.8.$$

Optimal hyperplane: $\boxed{-0.4x_1 + 1.2x_2 + 4.8 = 0}$.

(b) Test predictions

Rule: $\hat{y} = \text{sign}(\mathbf{w}^{*\top} \mathbf{x} + b^*) = \text{sign}(-0.4x_1 + 1.2x_2 + 4.8)$.

point	1	2	3	4	5	6	7	8
$f(\mathbf{x})$	3.2	2.8	4.8	2.6	−2.0	−1.0	−4.0	−3.2
$\hat{y}$	+1	+1	+1	+1	−1	−1	−1	−1

All 8 predictions are correct (100%).

(c) Removing points

Remove point 2 ($\alpha_2 = 0$, non-support vector): $\mathbf{w}, b$ unchanged $\Rightarrow$ predictions unchanged.
Remove point 8 (support vector): the solution changes ($\mathbf{w}^* \approx (0.235, 0.941)^\top$, $b^* \approx 2.059$), but all test predictions remain the same (still 100%).

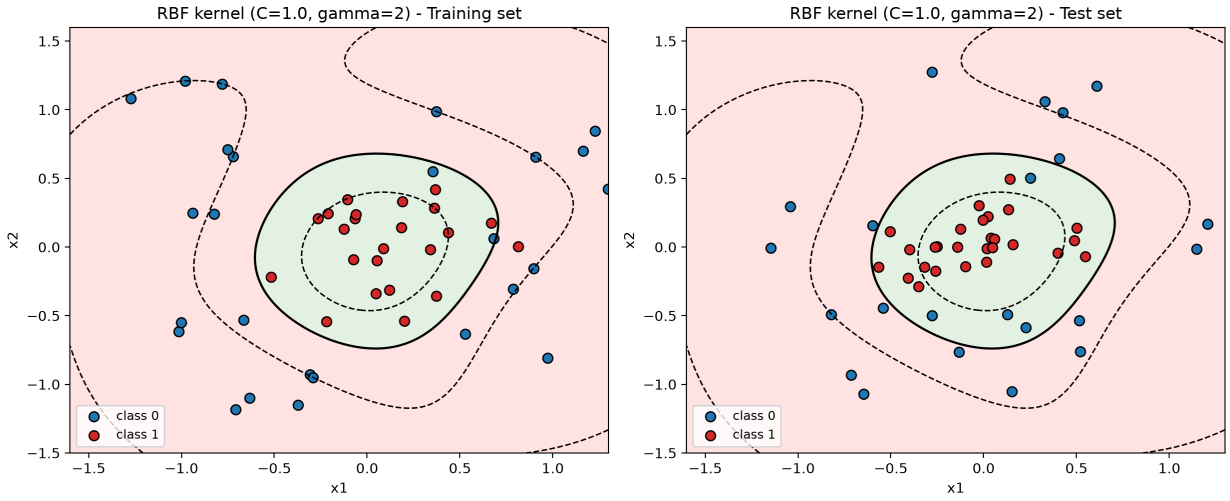
Question 2: Kernel SVM (scikit-learn)

- **Linear kernel,** $C = 1.0$: test accuracy 51.02% (matches the example).
- **RBF kernel,** $C = 1.0$, $\gamma = 2$: test accuracy **91.84**% (23 support vectors) — exceeds the required 85%.
- Also tried: polynomial (67.3%), sigmoid (46.9%).

Core code:

```
from sklearn.svm import SVC
clf = SVC(kernel='rbf', C=1.0, gamma=2)
clf.fit(X_train, y_train)
acc = clf.score(X_test, y_test)     # 0.9184
```

Plots (solid line = decision boundary, dashed lines = margins = ± 1):



Question 3: K-means and DBSCAN

Points: $a(1,2)$, $b(2,3)$, $c(2,4)$, $d(6,5)$, $e(3,1)$, $f(3,3)$, $g(4,4)$, $h(7,6)$, $i(8,8)$, $j(8,9)$, $k(9,4)$, $l(5,9)$, $m(4,7)$, $n(6,8)$, $o(6,4)$.

(a) K-means, first iteration

Initial centres: $\mu_1 = (7,4)$, $\mu_2 = (3,6)$, $\mu_3 = (7,9)$. Assignment by nearest centre: $\mu_1 : \{d, e, h, k, o\}$, $\mu_2 : \{a, b, c, f, g, m\}$, $\mu_3 : \{i, j, l, n\}$.

Updated centres:

$$\mu_1^{(1)} = \boxed{(6.2,\ 4)}, \qquad \mu_2^{(1)} = \boxed{(2.67,\ 3.83)}, \qquad \mu_3^{(1)} = \boxed{(6.75,\ 8.5)}.$$

q2_svm.py - Assignment2 - Visual Studio Code

q2_svm.py

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.svm import SVC

# ---------- load data (snippet from the assignment) ----------
train = pd.read_csv('training.csv', encoding='utf-8-sig')
test  = pd.read_csv('test.csv',  encoding='utf-8-sig', sep=r'[,\t]+', engine='python')
X_train, y_train = train.iloc[:, :-1].to_numpy().astype(float), train.iloc[:, -1].to_numpy().astype(float)
X_test,  y_test  = test.iloc[:, :-1].to_numpy().astype(float),  test.iloc[:, -1].to_numpy().astype(float)

# ---------- Model 1: linear kernel, soft margin ----------
clf_lin = SVC(kernel='linear', C=1.0)
clf_lin.fit(X_train, y_train)
acc_lin = clf_lin.score(X_test, y_test)
print(f'linear kernel, C=1.0:        test accuracy = {acc_lin:.4f}')

# ---------- Model 2: RBF kernel, soft margin ----------
clf_rbf = SVC(kernel='rbf', C=1.0, gamma=2)
clf_rbf.fit(X_train, y_train)
acc_rbf = clf_rbf.score(X_test, y_test)
print(f'RBF kernel, C=1.0, gamma=2:  test accuracy = {acc_rbf:.4f}')
print(f'number of support vectors = {len(clf_rbf.support_)}')

# ---------- 2D plot: decision boundary (solid) + margins (dashed) ----------
def plot_svm(ax, X, y, clf, title):
    x_min, x_max = -1.6, 1.3
    y_min, y_max = -1.5, 1.6
    xx, yy = np.meshgrid(np.linspace(x_min, x_max, 400),
                         np.linspace(y_min, y_max, 400))
    Z = clf.decision_function(np.c_[xx.ravel(), yy.ravel()]).reshape(xx.shape)
    ax.contourf(xx, yy, Z, levels=[-10, 0, 10], colors=['#ffd0d0', '#cfe8cf'], alpha=0.6)
    ax.contour(xx, yy, Z, levels=[0],  colors='k', linewidths=1.6, linestyles='solid')   # decision boundary
    ax.contour(xx, yy, Z, levels=[-1, 1], colors='k', linewidths=1.1, linestyles='dashed') # margins
    ax.scatter(X[y == 0, 0], X[y == 0, 1], c='#1f77b4', s=50, label='class 0', edgecolors='k')
    ax.scatter(X[y == 1, 0], X[y == 1, 1], c='#d62728', s=50, label='class 1', edgecolors='k')
    ax.set_xlabel('x1'); ax.set_ylabel('x2'); ax.set_title(title)
    ax.legend(loc='lower left', fontsize=9)

fig, axes = plt.subplots(1, 2, figsize=(12.5, 5.2))
plot_svm(axes[0], X_train, y_train, clf_rbf, 'RBF kernel (C=1.0, gamma=2) - Training set')
plot_svm(axes[1], X_test,  y_test,  clf_rbf, 'RBF kernel (C=1.0, gamma=2) - Test set')
plt.tight_layout()
plt.savefig('q2_rbf.png', dpi=200)
plt.show()
```

(b) DBSCAN: $\epsilon = 2$, minpts$= 3$

ϵ-neighbourhoods (point itself included):

a	$\{a, b\}$	i	$\{i, j, n\}$
b	$\{a, b, c, f\}$	j	$\{i, j\}$
c	$\{b, c, f, g\}$	k	$\{k\}$
d	$\{d, h, o\}$	l	$\{l, n\}$
e	$\{e, f\}$	m	$\{m\}$
f	$\{b, c, e, f, g\}$	n	$\{i, l, n\}$
g	$\{c, f, g, o\}$	o	$\{d, g, o\}$
h	$\{d, h\}$		

Core points ($|N_\epsilon| \geq 3$): b, c, d, f, g, i, n, o.
Border points: a, e, h, j, l.
Noise points: k, m.

(c) Is h density-reachable from b?

Yes, via the chain $\boxed{\text{b} \to c \to g \to o \to d \to h}$ (each step from a core point to a point in its ϵ-neighbourhood).

(d) Clusters

Cluster 1: $\{a, b, c, d, e, f, g, h, o\}$; Cluster 2: $\{i, j, l, n\}$; Noise: k, m.

Question 4: Hierarchical clustering (group average)

	A	B	C	D	E
A	0	8	6	2	1
B	8	0	7	3	6
C	6	7	0	9	5
D	2	3	9	0	4
E	1	6	5	4	0

Group average: $d(U,V) = \frac{1}{|U||V|} \sum_{u \in U} \sum_{v \in V} d(u,v)$.

Step 1. $d(A,E) = 1$ min $\Rightarrow$ merge into AE.

Updated: AE–$B = 7$, AE–$C = 5.5$, AE–$D = 3$.

Step 2. $d(B,D) = 3$ min $\Rightarrow$ merge into BD.

$d(AE,BD) = \frac{8+6+2+4}{4} = 5$, $d(C,BD) = \frac{7+9}{2} = 8$.

Step 3. $d(AE,BD) = 5$ min $\Rightarrow$ merge into $AEBD$.

$d(C,AEBD) = \frac{6+5+7+9}{4} = 6.75$.

Step 4. All merged at distance 6.75.

Merge order: $\boxed{(A,E) \xrightarrow{1} (B,D) \xrightarrow{3} (AE,BD) \xrightarrow{5} (AEBD,C) \xrightarrow{6.75}}$

